

SUSHANT LOKHANDE

San Jose, CA | +1 669-340-6655 | lokhandesushant014@gmail.com | sushantlokhande.me |
linkedin.com/in/sushantlokhande14 | github.com/sushantlokhande14

EXPERIENCE

San Jose State University | Graduate Research Assistant, Distributed Real-Time Systems Aug 2025 to May 2026

- Reduced rebuffering by **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configurations.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.
- Built and exercised **14 interoperability scenarios** covering networking, dependency, and **failure recovery** behavior in a distributed real-time system.
- Built and evaluated **RAG** pipelines (chunking, embeddings, vector retrieval, top-k, grounded prompts), separating **retrieval** failures from generation failures to guide fixes.

Turito | Software Engineering Intern, Real-Time Communications Jun 2022 to Oct 2022

- Worked across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**, including ICE/STUN/TURN connection setup and degraded-network handling.
- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Lowered network egress by **14%** by removing redundant media-path processing and simplifying routing and payload handling.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Surfaced backend fixes that raised equivalent-load signaling throughput by about **31%**.

TECHNICAL SKILLS

Languages: Java, Python, C++17, Rust, SQL, TypeScript

Real-Time & Networking: WebRTC, RTP/RTCP, ICE/STUN/TURN, QUIC, WebTransport, WebSockets

Backend & Platform: gRPC, Kafka, Temporal, FastAPI, REST, Redis, PostgreSQL, Docker, AWS, Linux

Reliability: retries and timeouts, load testing (Tokio), metrics, structured logging, pytest, CI/CD

PROJECTS

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Built an OpenAI-compatible gateway with exact and **semantic caching**, provider **failover**, and rate limiting that cut p50 latency from **759 ms to 44 ms** over a 4,000-request load test.

Portcullis, Policy Gateway | *Rust*

- Built a **Rust** proxy that centralizes authorization and rate limiting before requests reach services and emits auditable accept, reject, and throttle decisions.

TCP Sliding Window Simulator | *Python, raw sockets, Linux*

- Implemented handshake, sliding window, cumulative ACKs, and retransmission over raw sockets with live packet-state visualization.

OPEN SOURCE

- **corsairdev/corsair**: 6 merged **webhook signature verification** fixes (Telegram, Teams, SharePoint, Monday, GitLab, Jira) that fail closed
- **fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Distributed Computing, Advanced Algorithms, Database Systems, Computer Networks, Machine Learning, NoSQL

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)